

Nickname:

1.10 Quiz: Arithmetic and Geometric Sequences

A calculator is allowed. Show your working: write the formula you use, then substitute.

1. [Maximum mark: 8]

Consider the arithmetic sequence

$$15, 23, 31, 39, \dots$$

- (a) Write down the value of the common difference d . [1]
- (b) Find an expression for u_n , the n^{th} term. [2]
- (c) Find u_{30} . [2]
- (d) Show that 303 is a term of the sequence, and find which term it is. [3]

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2. [Maximum mark: 9]

In an arithmetic sequence, $u_5 = 38$ and $u_{14} = 11$.

- Find the common difference d and the first term u_1 . [3]
- Find the first term of the sequence that is negative. [3]

The first three terms of a different arithmetic sequence are

$$2k + 1, \quad 4k - 3, \quad 5k + 1.$$

- (c) Find the value of k , and write down the first three terms. [3]

(a) State whether each sequence is arithmetic, geometric or neither. Write down d or r where there is one. [4]

(d) Challenge: A different geometric sequence has $u_3 = 45$ and $u_6 = 1215$. Find the common ratio and the first term. [3]

Nickname:

4. [Maximum mark: 7]

Marco is offered two jobs.

Job A: \$38 000 in the first year, rising by \$2000 each year.

Job B: \$36 000 in the first year, rising by 6% each year.

(a) Write down an expression for the salary in year n for each job. [2]

(b) Find Marco's salary in year 3 for each job. [2]

(c) Find the first year in which the Job B salary is greater than the Job A salary. [3]

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